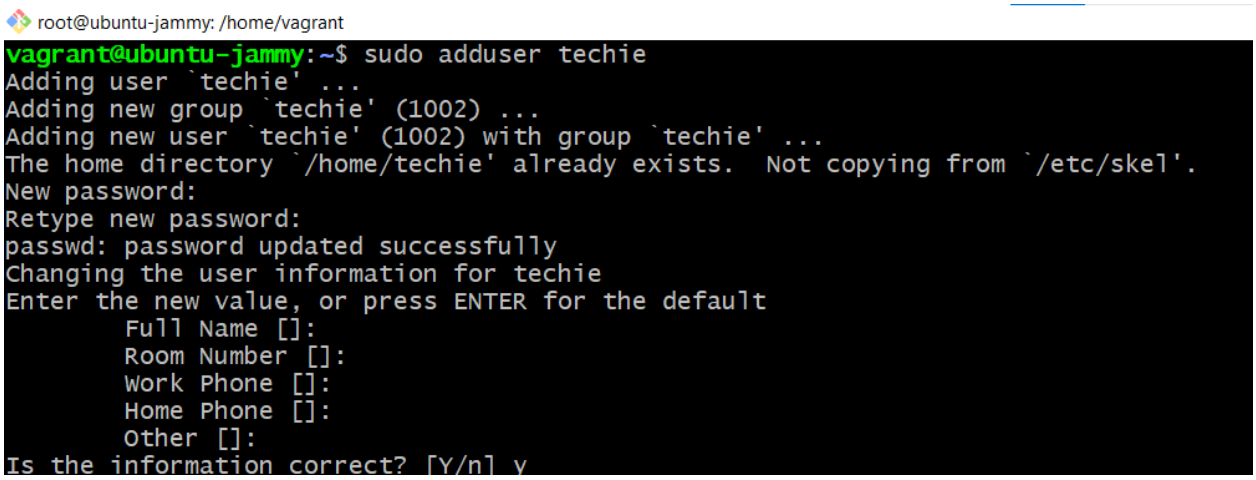
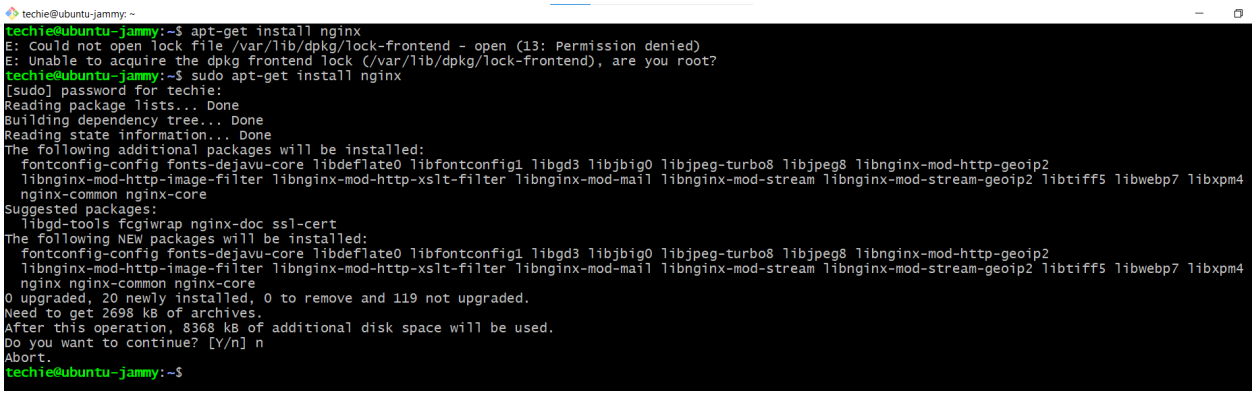
1. Create user with name Techie and provide sudo access to user.

Sudo adduser techie



- Open a file in etc/sudoers and provided techie user to sudo access.



```
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_pro

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
techie  ALL=(ALL:ALL) ALL
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
root@ubuntu-jammy:~#
```

2. Create a new directory.

Mkdir linux_task

```
techie@ubuntu-jammy: ~
techie@ubuntu-jammy:~$ mkdir linux_task
techie@ubuntu-jammy:~$ ls
linux_task
techie@ubuntu-jammy:~$
```

3. List the contents of a directory.

Touch creates blank files

techie@ubuntu-jammy: ~/linux_task

```
techie@ubuntu-jammy:~$ mkdir linux_task
techie@ubuntu-jammy:~$ ls
linux_task
techie@ubuntu-jammy:~$ cd linux_task/
techie@ubuntu-jammy:~/linux_task$ ls
techie@ubuntu-jammy:~/linux_task$ touch 1 2 3 4 5 a b c
techie@ubuntu-jammy:~/linux_task$ ls
1  2  3  4  5  a  b  c
techie@ubuntu-jammy:~/linux_task$
```

4. Change the current directory.

Cd for change directory

techie@ubuntu-jammy: ~/practise

```
techie@ubuntu-jammy:~/linux_task$ cd ..
techie@ubuntu-jammy:~$ mkdir practise
techie@ubuntu-jammy:~$ ls
linux_task  practise
techie@ubuntu-jammy:~$ cd practise/
techie@ubuntu-jammy:~/practise$
```

5. Create a new empty file.

Touch for new empty file

techie@ubuntu-jammy: ~/practise

```
techie@ubuntu-jammy:~/linux_task$ cd ..
techie@ubuntu-jammy:~$ mkdir practise
techie@ubuntu-jammy:~$ ls
linux_task  practise
techie@ubuntu-jammy:~$ cd practise/
techie@ubuntu-jammy:~/practise$ touch empty_file
techie@ubuntu-jammy:~/practise$ ls
empty_file
techie@ubuntu-jammy:~/practise$
```

6. View the contents of a file.

Cat empty_file

techie@ubuntu-jammy: ~/practise

```
techie@ubuntu-jammy:~/practise$ ls
empty_file
techie@ubuntu-jammy:~/practise$ echo "hi this is devops batch 23, my self sharikh khan" > empty_file
techie@ubuntu-jammy:~/practise$ cat empty_file
hi this is devops batch 23, my self sharikh khan
techie@ubuntu-jammy:~/practise$
```

7. Copy a file to another location.

cp empty_file /home/techie/linux_task/

techie@ubuntu-jammy: ~/linux_task

```
techie@ubuntu-jammy:~/practise$ ls
empty_file
techie@ubuntu-jammy:~/practise$ cp empty_file /home/techie/linux_task/
techie@ubuntu-jammy:~/practise$ cd /home/techie/linux_task/
techie@ubuntu-jammy:~/linux_task$ ls
1  2  3  4  5  a  b  c  empty_file
techie@ubuntu-jammy:~/linux_task$ cat empty_file
hi this is devops batch 23, my self sharikh khan
techie@ubuntu-jammy:~/linux_task$
```

8. Move a file to another location

mv linux_task/empty_file sharikh/

techie@ubuntu-jammy: ~/sharikh

```
techie@ubuntu-jammy:~$ mkdir sharikh
techie@ubuntu-jammy:~$ ls
linux_task  practise  sharikh
techie@ubuntu-jammy:~$ mv linux_task/empty_file sharikh/
techie@ubuntu-jammy:~$ cd linux_task/
techie@ubuntu-jammy:~/linux_task$ ls
1  2  3  4  5  a  b  c
techie@ubuntu-jammy:~/linux_task$ cd ..
techie@ubuntu-jammy:~$ cd sharikh/
techie@ubuntu-jammy:~/sharikh$ ls
empty_file
techie@ubuntu-jammy:~/sharikh$ at empty_file
Command 'at' not found, but can be installed with:
apt install at
Please ask your administrator.
techie@ubuntu-jammy:~/sharikh$ cat empty_file
hi this is devops batch 23, my self sharikh khan
techie@ubuntu-jammy:~/sharikh$
```

9. Rename a file.

mv empty_file blank_file

techie@ubuntu-jammy: ~/sharikh

```
techie@ubuntu-jammy:~/sharikh$ ls
empty_file
techie@ubuntu-jammy:~/sharikh$ mv empty_file blank_file
techie@ubuntu-jammy:~/sharikh$ ls
blank_file
techie@ubuntu-jammy:~/sharikh$
```

10. Delete a file

rm blank_file

techie@ubuntu-jammy: ~/sharikh

```
techie@ubuntu-jammy:~/sharikh$ ls
blank_file
techie@ubuntu-jammy:~/sharikh$ rm blank_file
techie@ubuntu-jammy:~/sharikh$ ls
techie@ubuntu-jammy:~/sharikh$
```

11. Grant or revoke permissions on a file or directory.

Chmod 777 empty_file read,write,execute for all.

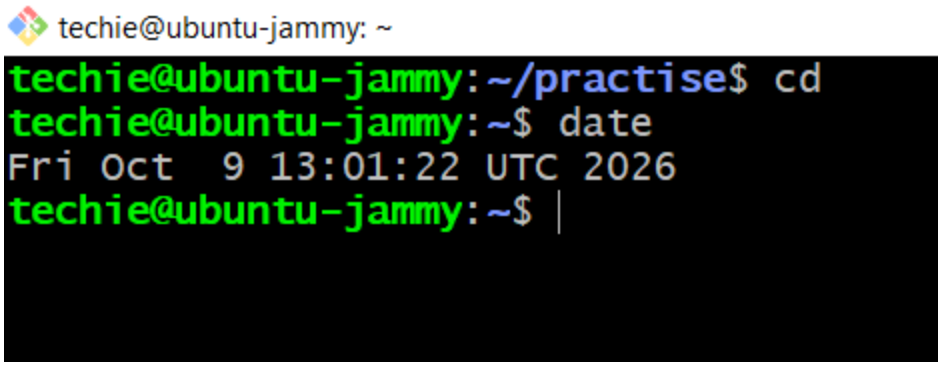
Chmod 641 empty_file red write for user ; read for group; execute for other.

techie@ubuntu-jammy: ~/practise

```
techie@ubuntu-jammy:~/practise$ ll
total 12
drwxrwxr-x 2 techie techie 4096 Oct  9 12:41 ./
drwxr-x--- 5 techie techie 4096 Oct  9 12:49 ../
-rw-rw-r-- 1 techie techie   49 Oct  9 12:43 empty_file
techie@ubuntu-jammy:~/practise$ chmod 777 empty_file
techie@ubuntu-jammy:~/practise$ ll
total 12
drwxrwxr-x 2 techie techie 4096 Oct  9 12:41 ./
drwxr-x--- 5 techie techie 4096 Oct  9 12:49 ../
-rwxrwxrwx 1 techie techie   49 Oct  9 12:43 empty_file*
techie@ubuntu-jammy:~/practise$ chmod 641 empty_file
techie@ubuntu-jammy:~/practise$ ll
total 12
drwxrwxr-x 2 techie techie 4096 Oct  9 12:41 ./
drwxr-x--- 5 techie techie 4096 Oct  9 12:49 ../
-rw-r----x 1 techie techie   49 Oct  9 12:43 empty_file*
techie@ubuntu-jammy:~/practise$
```

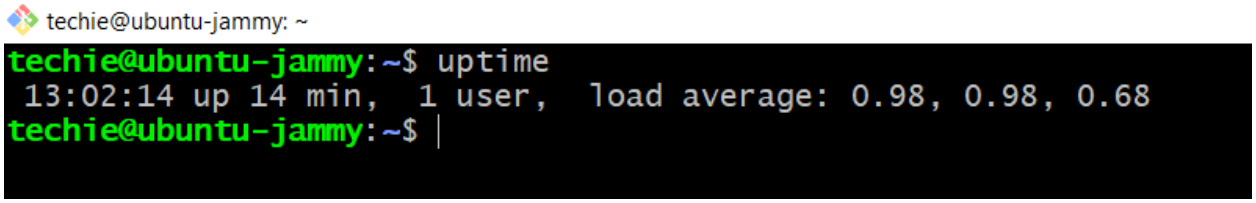
12. View the current date and time.

date



13. Check the system uptime.

uptime



14. View the running processes

Ps -ef

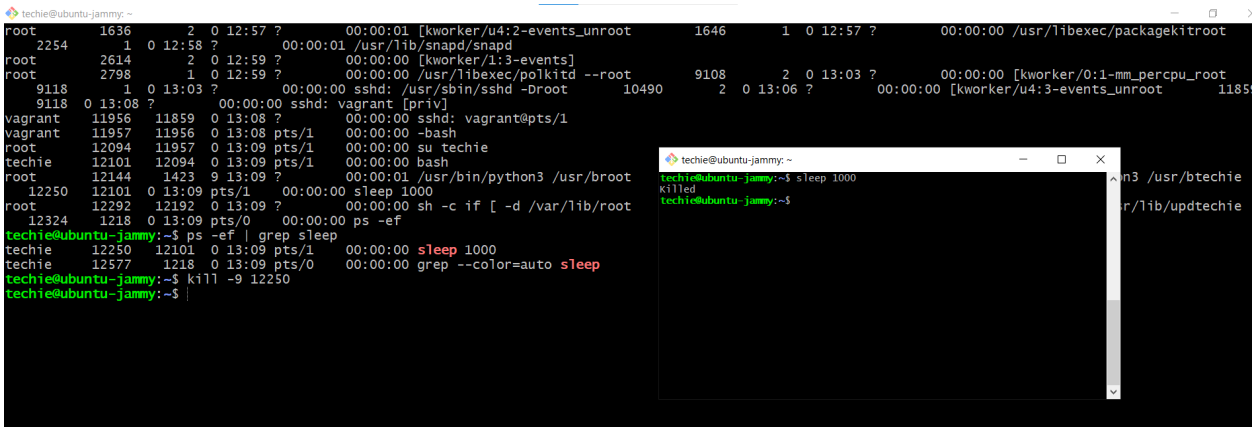
techie@ubuntu-jammy: ~

```
techie@ubuntu-jammy:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 12:47 ?        00:00:12 /sbin/init
root           2       0  0 12:47 ?        00:00:00 [kthreadd]
root           3       2  0 12:47 ?        00:00:00 [rcu_gp]
root           4       2  0 12:47 ?        00:00:00 [rcu_par_gp]
root           5       2  0 12:47 ?        00:00:00 [slub_flushwq]
root           6       2  0 12:47 ?        00:00:00 [netns]
root           7       2  0 12:47 ?        00:00:01 [kworker/0:0-events]
root           8       2  0 12:47 ?        00:00:00 [kworker/0:0H-events_highpri]
root           9       2  0 12:47 ?        00:00:02 [kworker/u4:0-events_unbound]
root          10       2  0 12:47 ?        00:00:00 [mm_percpu_wq]
root          11       2  0 12:47 ?        00:00:00 [rcu_tasks_rude_]
root          12       2  0 12:47 ?        00:00:00 [rcu_tasks_trace]
root          13       2  0 12:47 ?        00:00:00 [ksoftirqd/0]
root          14       2  0 12:47 ?        00:00:01 [rcu_sched]
root          15       2  0 12:47 ?        00:00:00 [migration/0]
root          16       2  0 12:47 ?        00:00:00 [idle_inject/0]
root          17       2  0 12:47 ?        00:00:00 [kworker/0:1-cgroup_release]
root          18       2  0 12:47 ?        00:00:00 [cpuhp/0]
root          19       2  0 12:47 ?        00:00:00 [cpuhp/1]
root          20       2  0 12:47 ?        00:00:00 [idle_inject/1]
root          21       2  0 12:47 ?        00:00:00 [migration/1]
root          22       2  0 12:47 ?        00:00:00 [ksoftirqd/1]
root          23       2  0 12:47 ?        00:00:00 [kworker/1:0-events]
root          24       2  0 12:47 ?        00:00:00 [kworker/1:0H-events_highpri]
root          25       2  0 12:47 ?        00:00:00 [kdevtmpfs]
```

15. Kill a running process.

Created dummy process using sleep 1000

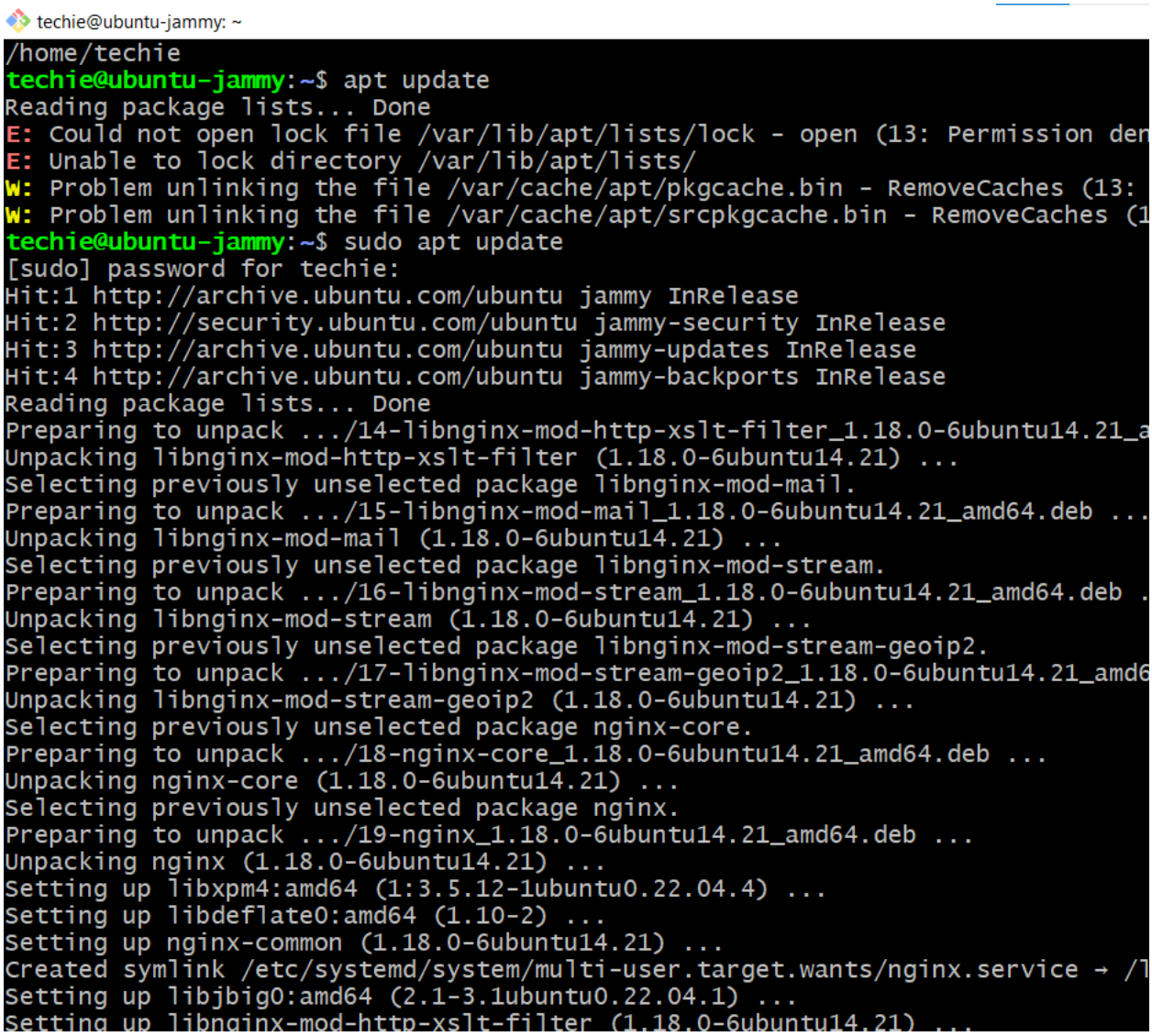
Kill -9 PID

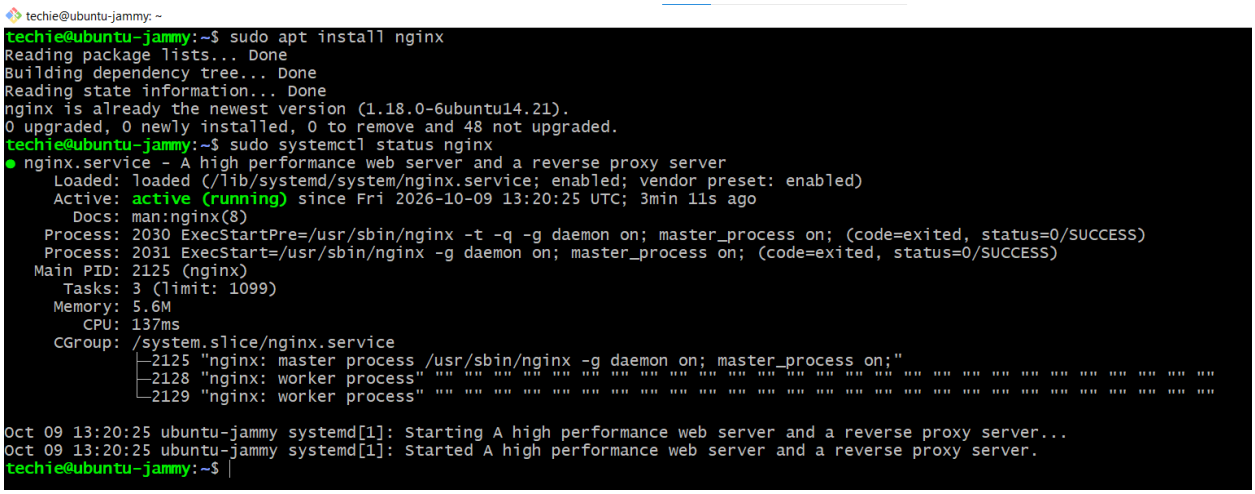


16. Install a package using the package manager (e.g., apt or yum).

First update sudo apt update

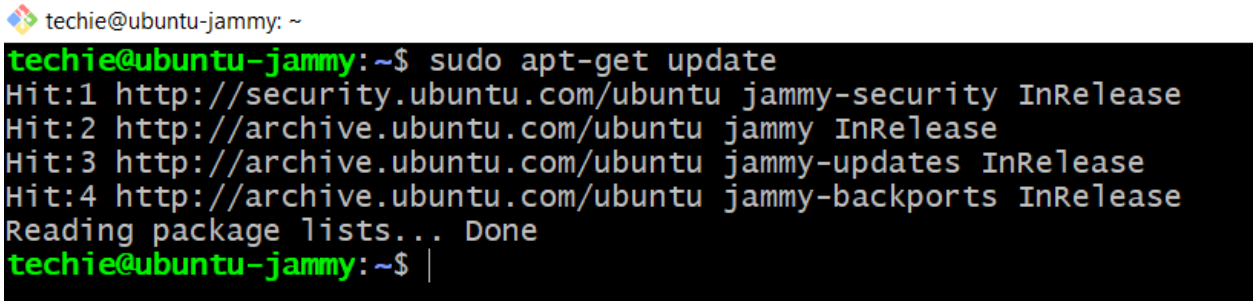
Sudo apt install ngnix





17. Update the system packages.

Sudo apt-get update



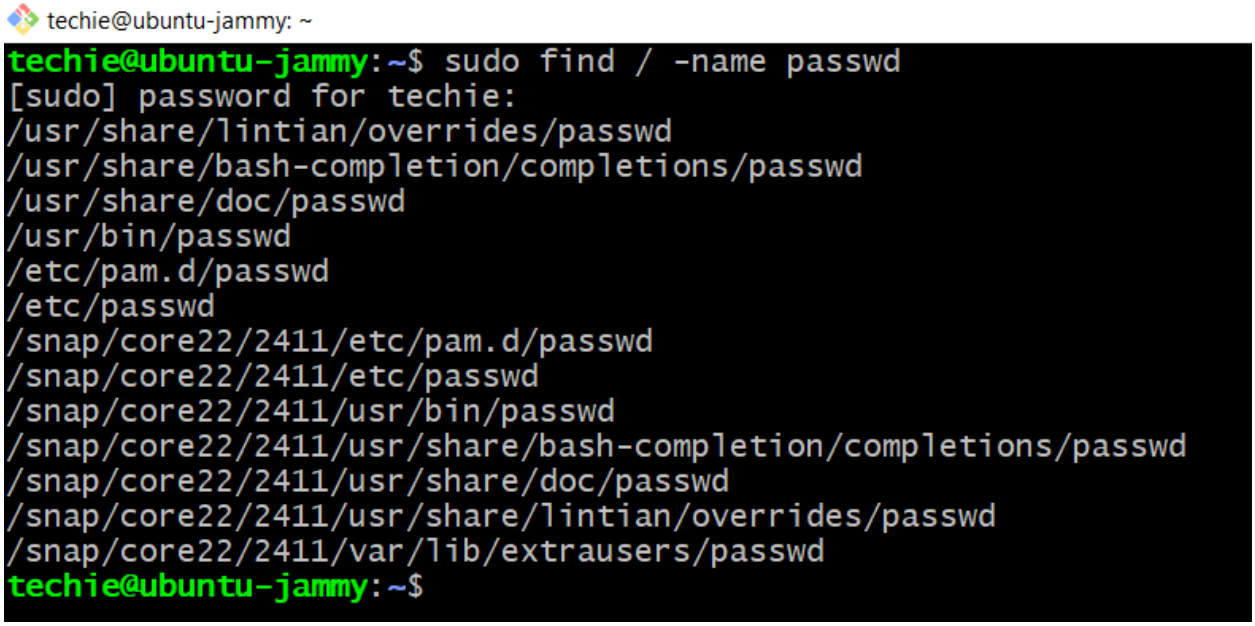
18. Create a symbolic link.

ln -s /home/techie/practise/empty_file empty_file_link



19. search for files using the find command.

Sudo find / -name <file name>



20. Compress and decompress files using tar

Compress using - tar -cvf backup.tar.gz /file_name

Decompress using - tar -xvf backup.tar.gz /file_name

techie@ubuntu-jammy: ~/practise

```
techie@ubuntu-jammy:~/sharikh$ touch devops linux ubuntu
techie@ubuntu-jammy:~/sharikh$ mkdir 1 2 3
techie@ubuntu-jammy:~/sharikh$ ls
1  2  3  devops  linux  ubuntu
techie@ubuntu-jammy:~/sharikh$ mkdir com
techie@ubuntu-jammy:~/sharikh$ ls
1  2  3  com  devops  linux  ubuntu
techie@ubuntu-jammy:~/sharikh$ cd ..
techie@ubuntu-jammy:~$ ls
practise  sharikh
techie@ubuntu-jammy:~$ tar -cvf sharikh/ backup_sharikh.tar
tar: sharikh/: Cannot open: Is a directory
tar: Error is not recoverable: exiting now
techie@ubuntu-jammy:~$ tar -cvf backup_sharikh.tar.gz /s
sbin/ snap/ srv/  sys/
techie@ubuntu-jammy:~$ tar -cvf backup_sharikh.tar.gz sharikh/
sharikh/
sharikh/1/
sharikh/linux
sharikh/3/
sharikh/2/
sharikh/com/
sharikh/ubuntu
sharikh/devops
techie@ubuntu-jammy:~$ ls
backup_sharikh.tar.gz  practise  sharikh
techie@ubuntu-jammy:~$ cp backup_sharikh.tar.gz practise/
techie@ubuntu-jammy:~$ cd practise/
techie@ubuntu-jammy:~/practise$ ls
backup_sharikh.tar.gz  empty_file
techie@ubuntu-jammy:~/practise$ tar -xvf backup_sharikh.tar.gz
sharikh/
sharikh/1/
sharikh/linux
sharikh/3/
sharikh/2/
sharikh/com/
sharikh/ubuntu
sharikh/devops
techie@ubuntu-jammy:~/practise$ ls
backup_sharikh.tar.gz  empty_file  sharikh
techie@ubuntu-jammy:~/practise$
```

21. Monitor system resources with top or htop

techie@ubuntu-jammy: ~

```
top - 13:51:17 up 34 min,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 109 total,   1 running, 108 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :    957.2 total,     91.5 free,    170.7 used,    695.0 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    625.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   3015 techie    20   0   10912   3812   3208 R   0.7   0.4   0:00.05 top
   2720 root      20   0       0      0      0 I   0.3   0.0   0:01.69 kworker/1:2-mpt_poll_0
      1 root      20   0  167536  12840   8248 S   0.0   1.3   0:07.95 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_rude_
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_trace
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.16 ksoftirqd/0
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.46 rcu_sched
     15 root      rt   0       0      0      0 S   0.0   0.0   0:00.02 migration/0
     16 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.66 migration/1
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.27 ksoftirqd/1
     23 root      20   0       0      0      0 I   0.0   0.0   0:01.28 kworker/1:0-events
     24 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     25 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     26 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     28 root      20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     30 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
     31 root      20   0       0      0      0 S   0.0   0.0   0:00.36 kcompactd0
     32 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     33 root      39  19       0      0      0 S   0.0   0.0   0:00.00 khugepaged
```

vagrant@ubuntu-jammy: ~

```
  0[                                                          0.0%]   Tasks: 39, 24 thr; 1 running
  1[|||                                                       3.4%]   Load average: 0.10 0.03 0.01
Mem[|||||||||||||||||||||||||||||||||||||||||||||||||||||196M/957M]   Uptime: 00:35:57
Swp[                                                        0K/0K]

    PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
   3091 vagrant    20   0  8704  4332  3276 R  2.7  0.4  0:00.18 htop
      1 root       20   0  163M 12840  8248 S  0.0  1.3  0:07.97 /sbin/init
    366 root       19  -1 72712 26192 25068 S  0.0  2.7  0:00.81 /lib/systemd/systemd-journald
    399 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.79 /sbin/multipathd -d -s
    402 root       20   0 22788  6268  4692 S  0.0  0.6  0:00.72 /lib/systemd/systemd-udevd
    404 root       20   0  346M 27228  9072 S  0.0  2.8  0:00.00 /sbin/multipathd -d -s
    405 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.00 /sbin/multipathd -d -s
    406 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.00 /sbin/multipathd -d -s
    407 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.02 /sbin/multipathd -d -s
    408 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.35 /sbin/multipathd -d -s
    409 root       RT   0  346M 27228  9072 S  0.0  2.8  0:00.06 /sbin/multipathd -d -s
    439 systemd-t  20   0 89364  6628  5824 S  0.0  0.7  0:00.44 /lib/systemd/systemd-timesyncd
    465 systemd-t  20   0 89364  6628  5824 S  0.0  0.7  0:00.00 /lib/systemd/systemd-timesyncd
    562 systemd-n  20   0 16136  8140  7128 S  0.0  0.8  0:00.25 /lib/systemd/systemd-networkd
    566 systemd-r  20   0 25680 12572  8256 S  0.0  1.3  0:00.52 /lib/systemd/systemd-resolved
    602 messagebu  20   0  8720  4524  3960 S  0.0  0.5  0:00.72 @dbus-daemon --system --address=systemd: --nofork -
    607 root       20   0 82704  2024  1768 S  0.0  0.2  0:00.34 /usr/sbin/irqbalance --foreground
    608 root       20   0 33084 19044 10200 S  0.0  1.9  0:00.35 /usr/bin/python3 /usr/bin/networkd-dispatcher --run
    609 syslog     20   0  217M  4868  4044 S  0.0  0.5  0:00.09 /usr/sbin/rsyslogd -n -iNONE
    612 root       20   0 1296M 37576 24732 S  0.0  3.8  0:02.93 /usr/lib/snapd/snapd
    614 root       20   0 23712  7400  6396 S  0.0  0.8  0:00.50 /lib/systemd/systemd-logind
    618 root       20   0  7288  2844  2620 S  0.0  0.3  0:00.01 /usr/sbin/cron -f -P
    620 syslog     20   0  217M  4868  4044 S  0.0  0.5  0:00.03 /usr/sbin/rsyslogd -n -iNONE
    621 syslog     20   0  217M  4868  4044 S  0.0  0.5  0:00.00 /usr/sbin/rsyslogd -n -iNONE
    622 syslog     20   0  217M  4868  4044 S  0.0  0.5  0:00.02 /usr/sbin/rsyslogd -n -iNONE
    637 root       20   0 1296M 37576 24732 S  0.0  3.8  0:00.17 /usr/lib/snapd/snapd
    639 root       20   0  6220  1176  1088 S  0.0  0.1  0:00.01 /sbin/agetty -o -p -- \u --keep-baud 115200,57600,3
    641 root       20   0 1296M 37576 24732 S  0.0  3.8  0:00.00 /usr/lib/snapd/snapd
    642 root       20   0 1296M 37576 24732 S  0.0  3.8  0:00.56 /usr/lib/snapd/snapd
```

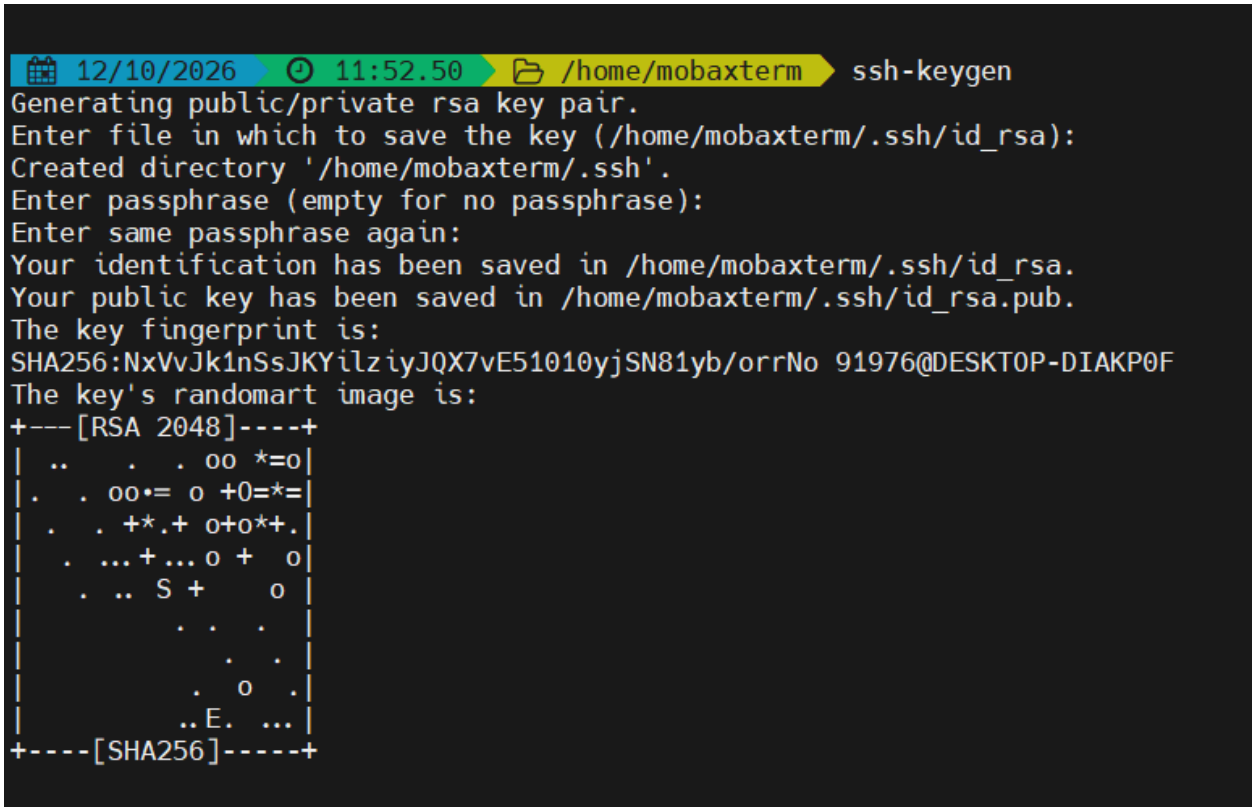
22. Create and manage user groups.

techie@ubuntu-jammy: ~

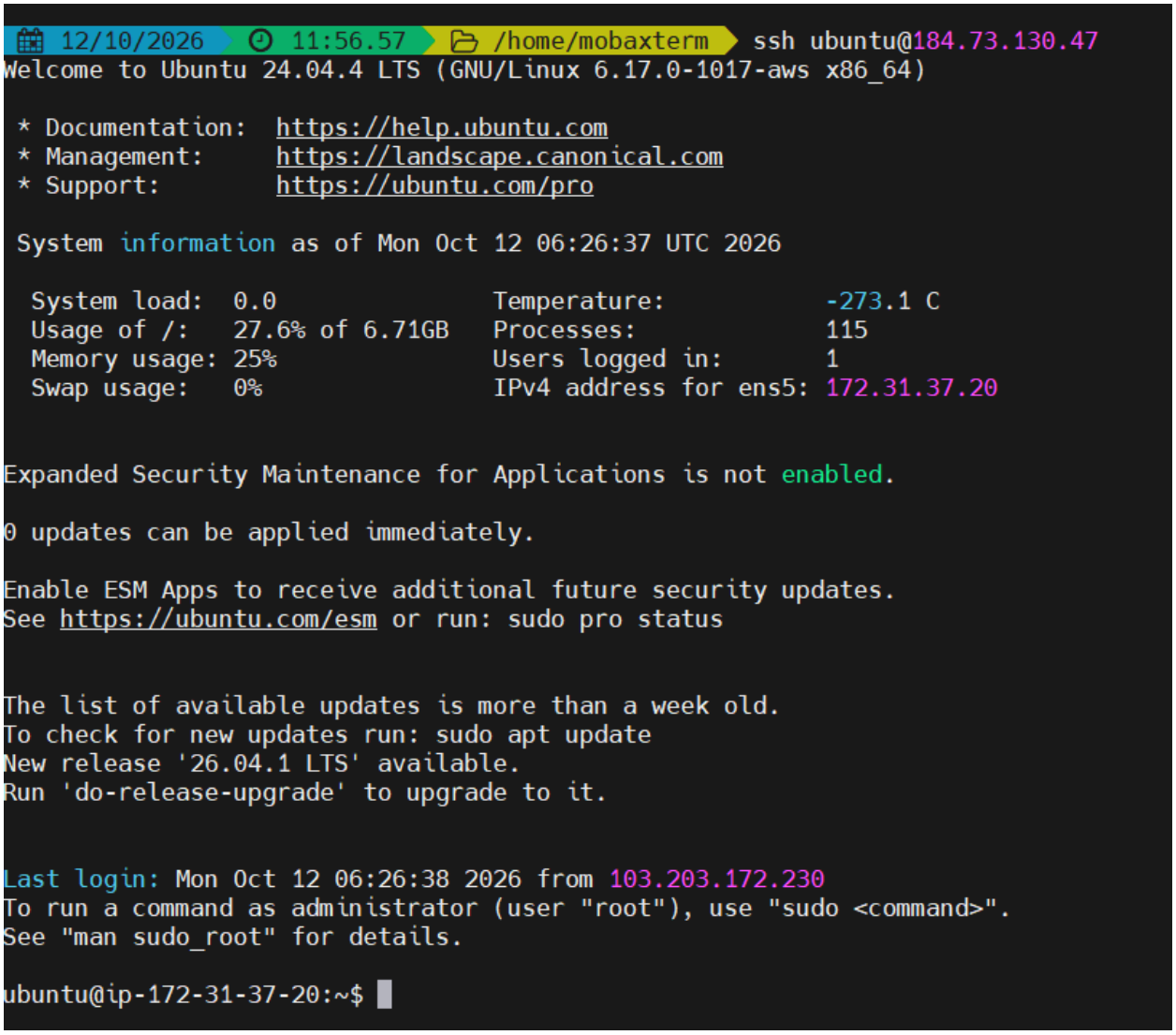
```
    default:
    default: Guest Additions Version: 6.0.0 r127566
    default: VirtualBox Version: 7.2
techie@ubuntu-jammy:~$ sudo groupadd techie_group
[sudo] password for techie:
techie@ubuntu-jammy:~$ sudo gpasswd techie_group
Changing the password for group techie_group
New Password:
Re-enter new password:
techie@ubuntu-jammy:~$ sudo usermod -aG techie_group techie
techie@ubuntu-jammy:~$ groups techie
techie : techie techie_group
techie@ubuntu-jammy:~$
```

23. Set up SSH password less authentication.

I have generated private and public key using ssh-keygen then i have copied public key and paste it in authorized key file in my ubuntu machine then i successfully loginto my remote server.







24. Monitor log files using tail or grep.

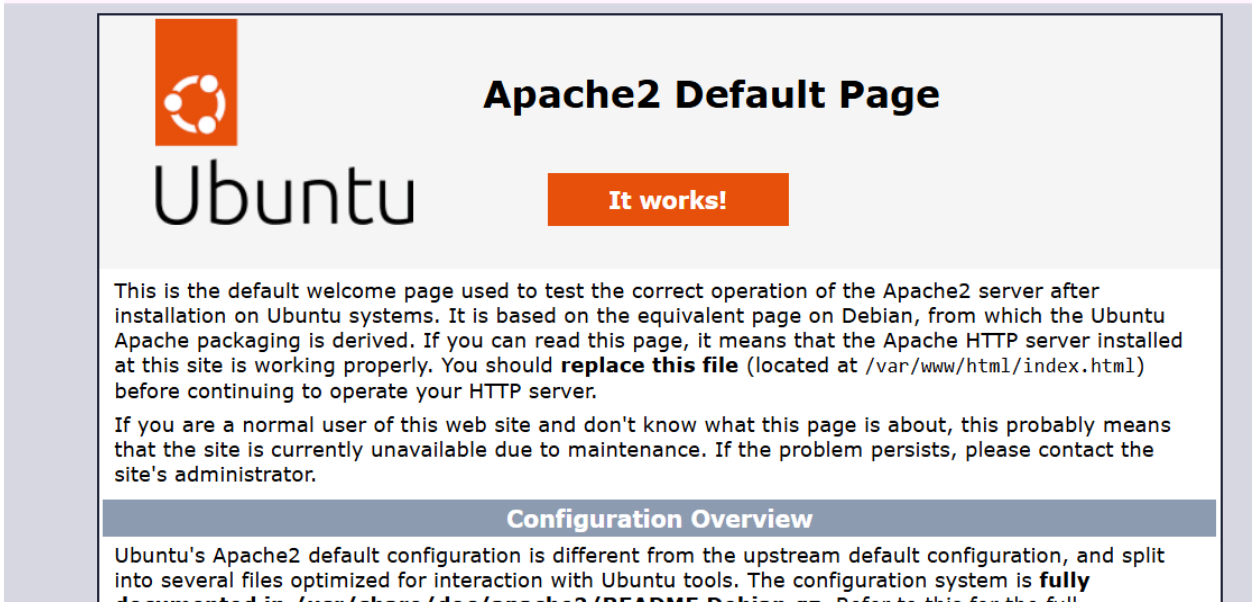
In one terminal i entered command tail -f syslog and other terminal i tried to update and then continuously get logs updated on my second terminal

```
alternatives.log  apport.log  auth.log  chrony  cloud-init.log        dmesg        journal   landscap
root@ip-172-31-37-20:/var/log# tail -f syslog
2026-10-12T06:33:54.796577+00:00 ip-172-31-37-20 systemd[1]: session-15.scope: Deactivated successfully
2026-10-12T06:35:01.356255+00:00 ip-172-31-37-20 CRON[1715]: (root) CMD (command -v debian-sa1 > /dev/n
2026-10-12T06:35:34.643139+00:00 ip-172-31-37-20 kernel: NOTICE: Automounting of tracing to debugfs is
2026-10-12T06:40:51.322969+00:00 ip-172-31-37-20 systemd[1]: Starting sysstat-collect.service - system
2026-10-12T06:40:51.326598+00:00 ip-172-31-37-20 systemd[1]: sysstat-collect.service: Deactivated succe
2026-10-12T06:40:51.326840+00:00 ip-172-31-37-20 systemd[1]: Finished sysstat-collect.service - system
2026-10-12T06:40:56.039878+00:00 ip-172-31-37-20 systemd[1]: Started session-17.scope - Session 17 of U
2026-10-12T06:40:58.717738+00:00 ip-172-31-37-20 systemd[1]: Started session-18.scope - Session 18 of U
2026-10-12T06:41:06.167473+00:00 ip-172-31-37-20 irqbalance[589]: Cannot change IRQ 28 affinity: Permis
2026-10-12T06:41:06.167746+00:00 ip-172-31-37-20 irqbalance[589]: IRQ 28 affinity is now unmanaged
2026-10-12T06:41:23.069099+00:00 ip-172-31-37-20 systemd[1]: Starting apt-news.service - Update APT New
2026-10-12T06:41:23.074183+00:00 ip-172-31-37-20 systemd[1]: Starting esm-cache.service - Update the lo
2026-10-12T06:41:23.432741+00:00 ip-172-31-37-20 python3[1933]: ["2026-10-12T06:41:23.432", "INFO", "ub
uld not open /proc/sysinfo", {}]
2026-10-12T06:41:23.433306+00:00 ip-172-31-37-20 python3[1933]: ["2026-10-12T06:41:23.432", "INFO", "ub
uld not open /sys/firmware/devicetree/base/model", {}]
2026-10-12T06:41:23.647146+00:00 ip-172-31-37-20 kernel: audit: type=1400 audit(1791787283.644:126): ap
="cap" profile="ubuntu_pro_esm_cache_systemd_detect_virt" pid=2046 comm="systemd-detect-" capability=38
2026-10-12T06:41:23.852191+00:00 ip-172-31-37-20 systemd[1]: apt-news.service: Deactivated successfully
2026-10-12T06:41:23.853506+00:00 ip-172-31-37-20 systemd[1]: Finished apt-news.service - Update APT New
2026-10-12T06:41:25.620777+00:00 ip-172-31-37-20 systemd[1]: esm-cache.service: Deactivated successfull
2026-10-12T06:41:25.622143+00:00 ip-172-31-37-20 systemd[1]: Finished esm-cache.service - Update the lo
2026-10-12T06:41:29.164876+00:00 ip-172-31-37-20 dbus-daemon[583]: [system] Activating via systemd: ser
it='packagekit.service' requested by ':1.35' (uid=0 pid=2168 comm="/usr/bin/gdbus call --system --dest
2026-10-12T06:41:29.170943+00:00 ip-172-31-37-20 systemd[1]: Starting packagekit.service - PackageKit D
2026-10-12T06:41:29.184619+00:00 ip-172-31-37-20 PackageKit: daemon start
2026-10-12T06:41:29.263543+00:00 ip-172-31-37-20 dbus-daemon[583]: [system] Successfully activated serv
2026-10-12T06:41:29.264074+00:00 ip-172-31-37-20 systemd[1]: Started packagekit.service - PackageKit Da
```

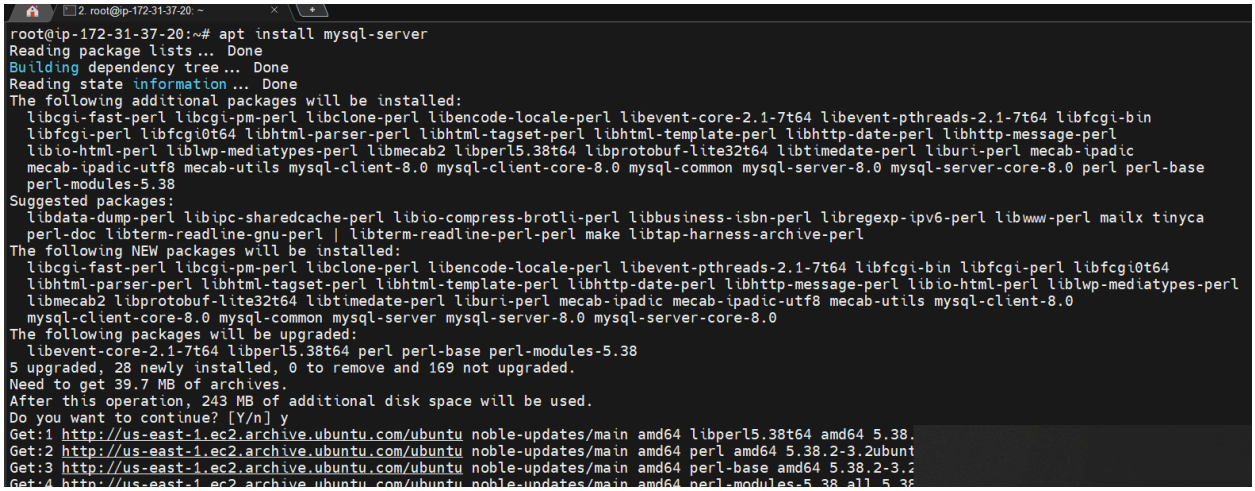
25. Set up a web server (e.g., Apache or Nginx).

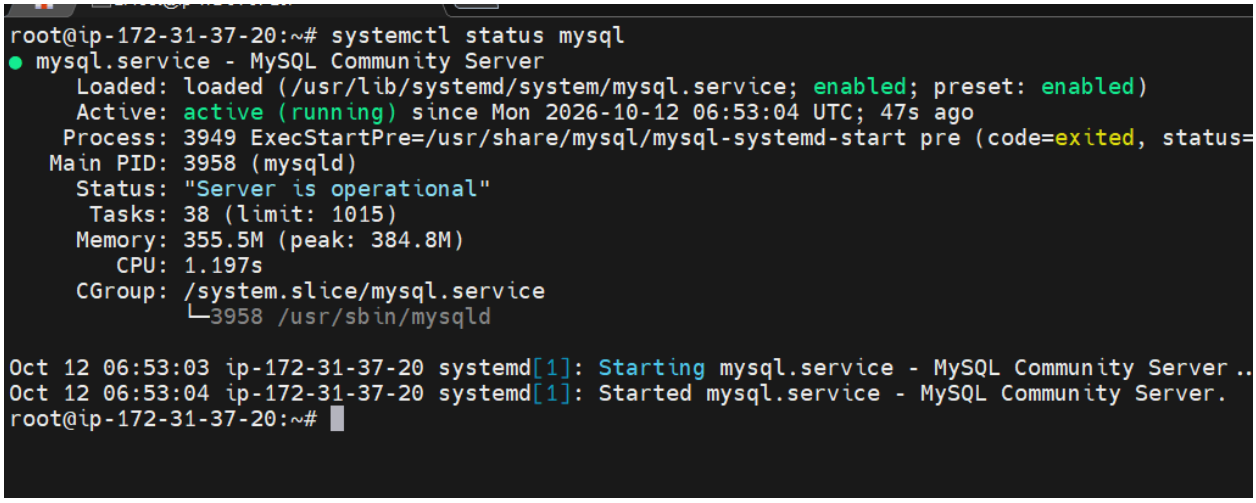
```
root@ip-172-31-37-20:~# apt install apache2
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 174 not upgraded.
Need to get 2096 kB of archives.
After this operation, 8129 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7.1 [92.7 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7.1 [11.2 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7.1 [9120 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.16 [1338 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.16 [163 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.16 [99.9 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.16 [90.2 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2096 kB in 0s (35.6 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 72608 files and directories currently installed.)
```

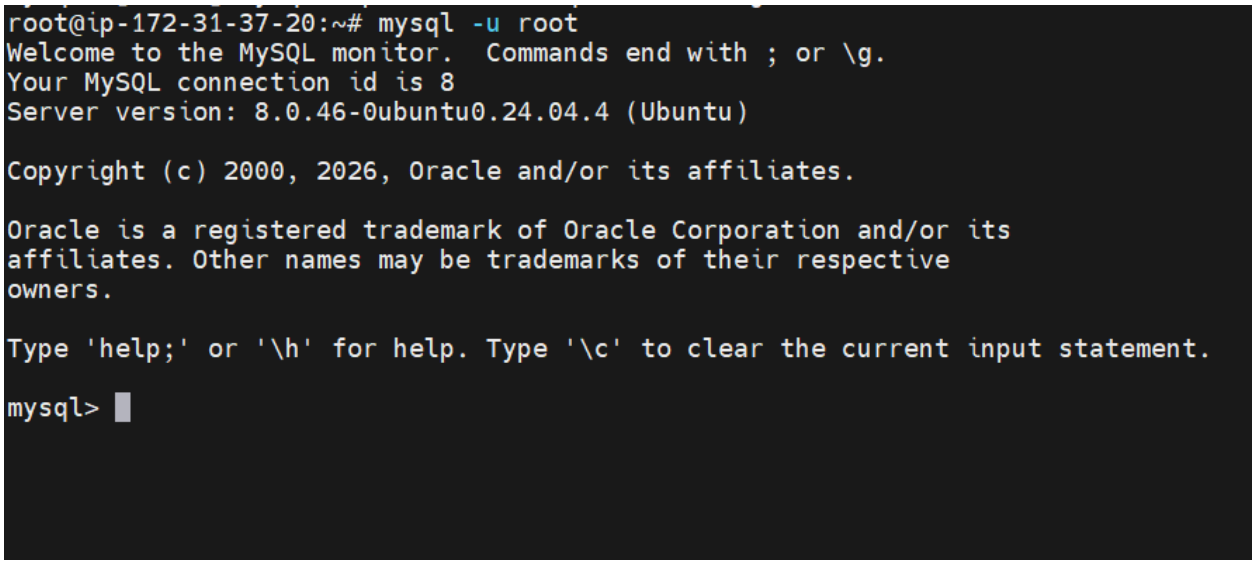
47



26. Configure and secure a MySQL Database



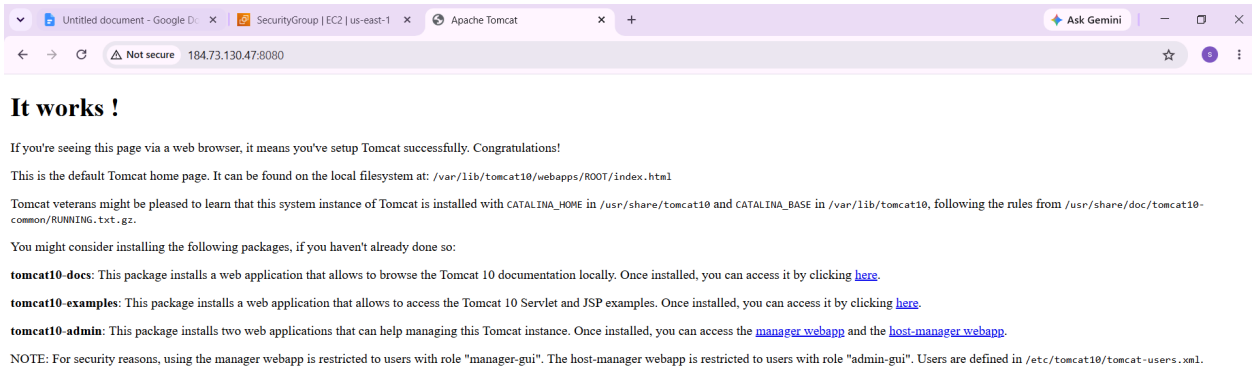




27. Set up a Application Server

Installed tomcat by using command

Apt install tomcat10 -y



28. create a service file for Apache Tomcat.(Should execute by using systemtctl command)

Service file for tomcat is already generated

```
oot@ip-172-31-37-20:~# sysctemctl cat tomcat10
ommand 'sysctemctl' not found, did you mean:
 command 'systemctl' from deb systemd (255.4-1ubuntu8.16)
 command 'systemctl' from deb systemctl (1.4.4181-1.1)
ry: apt install <deb name>
oot@ip-172-31-37-20:~# systemctl cat tomcat10
 /usr/lib/systemd/system/tomcat10.service

 Systemd unit file for Apache Tomcat


Unit]
escription=Apache Tomcat 10 Web Application Server
ocumentation=https://tomcat.apache.org/tomcat-10.0-doc/index.
fter=network.target
equiresMountsFor=/var/log/tomcat10 /var/lib/tomcat10

Service]

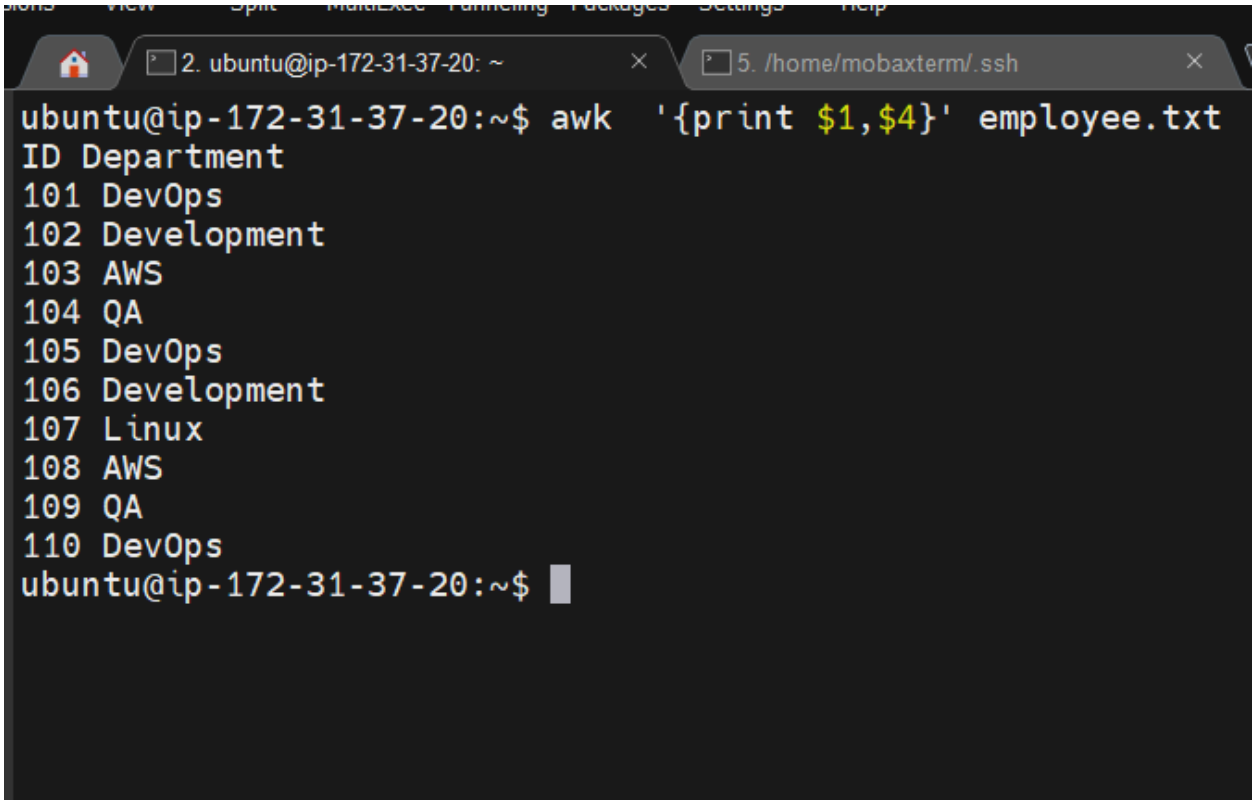
 Configuration
nvironment="CATALINA_HOME=/usr/share/tomcat10"
nvironment="CATALINA_BASE=/var/lib/tomcat10"
nvironment="CATALINA_TMPDIR=/tmp"
nvironment="JAVA_OPTS=-Djava.awt.headless=true"

 Lifecycle
ype=simple
xecStartPre=+/usr/libexec/tomcat10/tomcat-update-policy.sh
xecStart=/bin/sh /usr/libexec/tomcat10/tomcat-start.sh
uccessExitStatus=143
estart=on-abort

 Logging
yslogIdentifier=tomcat10

 Security
ser=tomcat
roup=tomcat
rivateTmp=yes
mbientCapabilities=CAP_NET_BIND_SERVICE
```

29. Print specific columns from a delimited file.



30. Filter and print lines based on a specific pattern or condition.

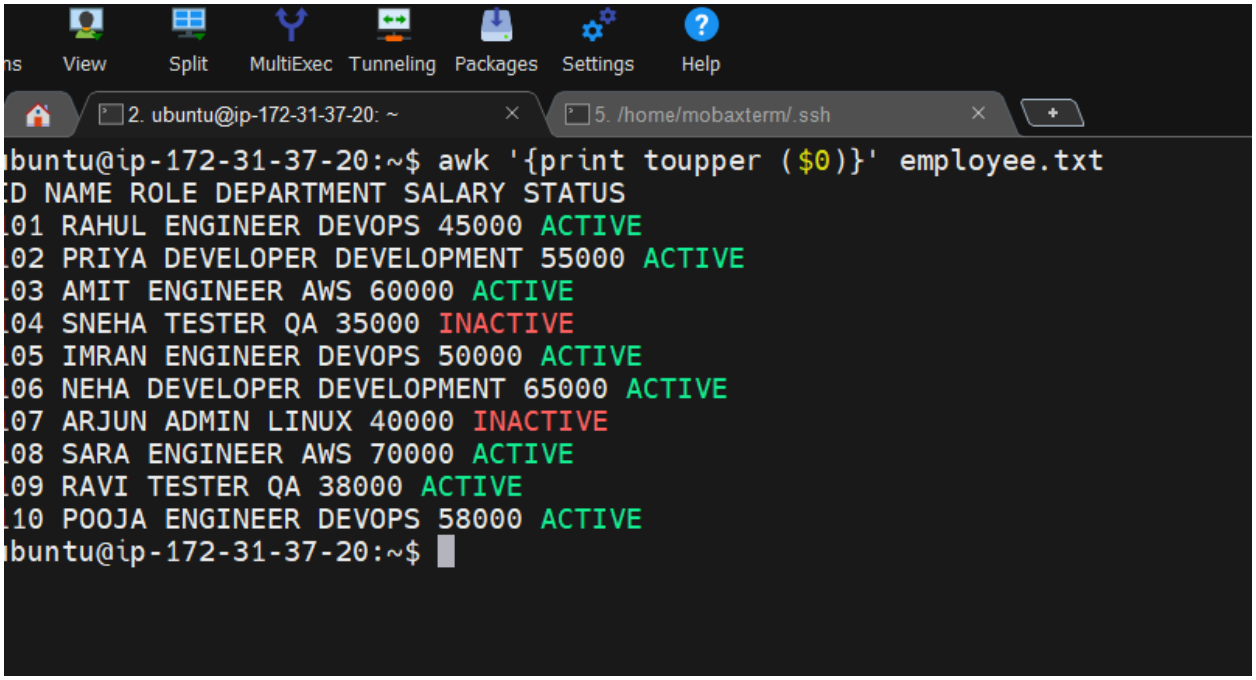
```
ubuntu@ip-172-31-37-20:~$ cat employee.txt
ID Name Role Department Salary Status
101 Rahul Engineer DevOps 45000 Active
102 Priya Developer Development 55000 Active
103 Amit Engineer AWS 60000 Active
104 Sneha Tester QA 35000 Inactive
105 Imran Engineer DevOps 50000 Active
106 Neha Developer Development 65000 Active
107 Arjun Admin Linux 40000 Inactive
108 Sara Engineer AWS 70000 Active
109 Ravi Tester QA 38000 Active
110 Pooja Engineer DevOps 58000 Active
ubuntu@ip-172-31-37-20:~$ awk /DevOps/'{print $1,$2,$4}' employee.txt
101 Rahul DevOps
105 Imran DevOps
110 Pooja DevOps
ubuntu@ip-172-31-37-20:~$
```

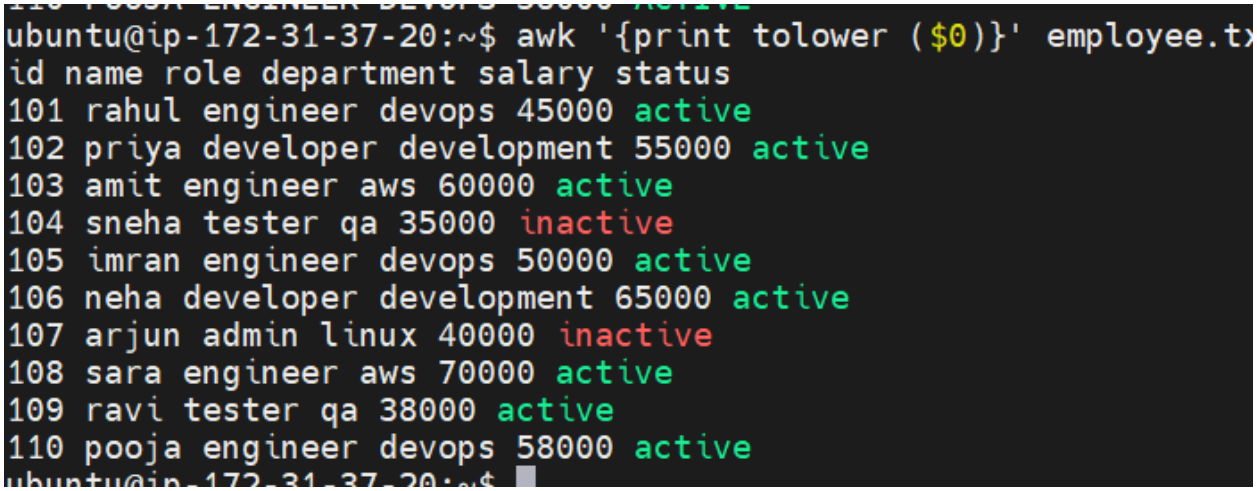
31. Calculate and print the average, sum, or other statistics of a column.

```
ubuntu@ip-172-31-37-20:~$ awk 'NR>1 {sum += $5} END {print "Total Salary:", sum}' employee.txt
Total Salary: 516000
ubuntu@ip-172-31-37-20:~$
```

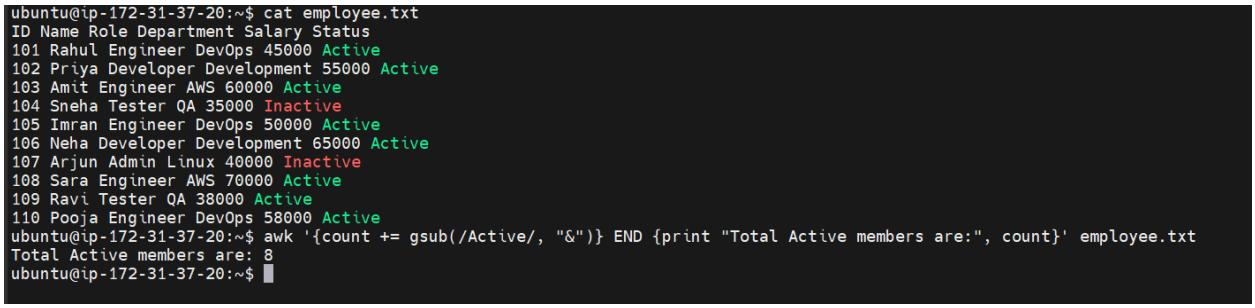
```
ubuntu@ip-172-31-37-20:~$ awk 'NR>1 {sum += $5; count++} END {print "Average Salary:", sum/count}' employee.txt
Average Salary: 51600
ubuntu@ip-172-31-37-20:~$
```

32. Perform string manipulation, such as extracting substrings or changing case

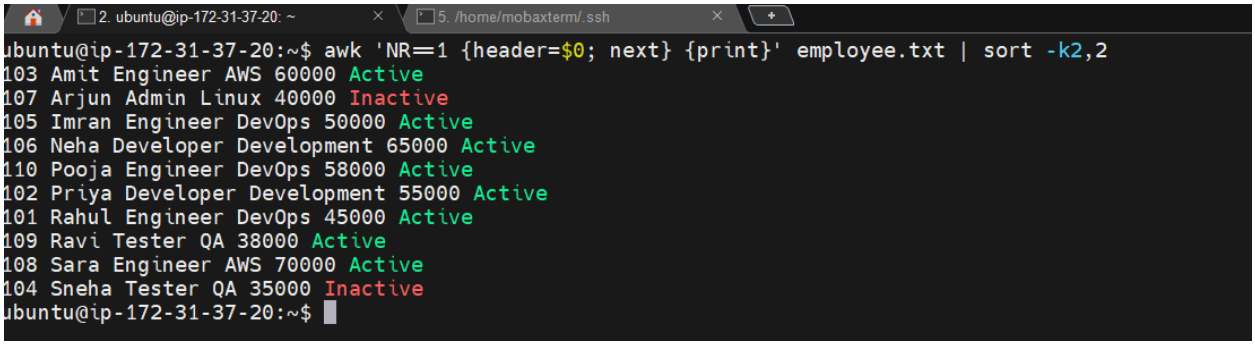




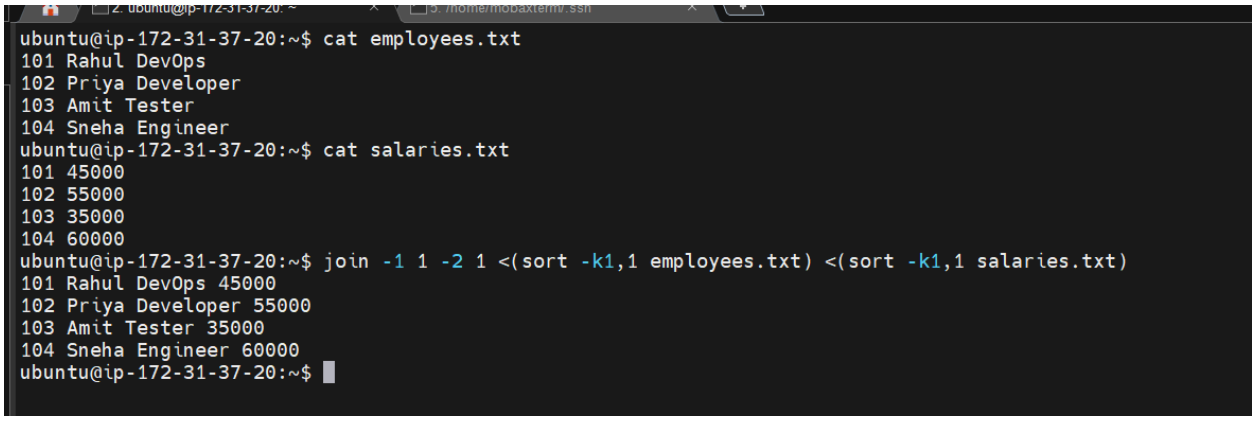
33. Count the occurrences of a specific pattern in a file.



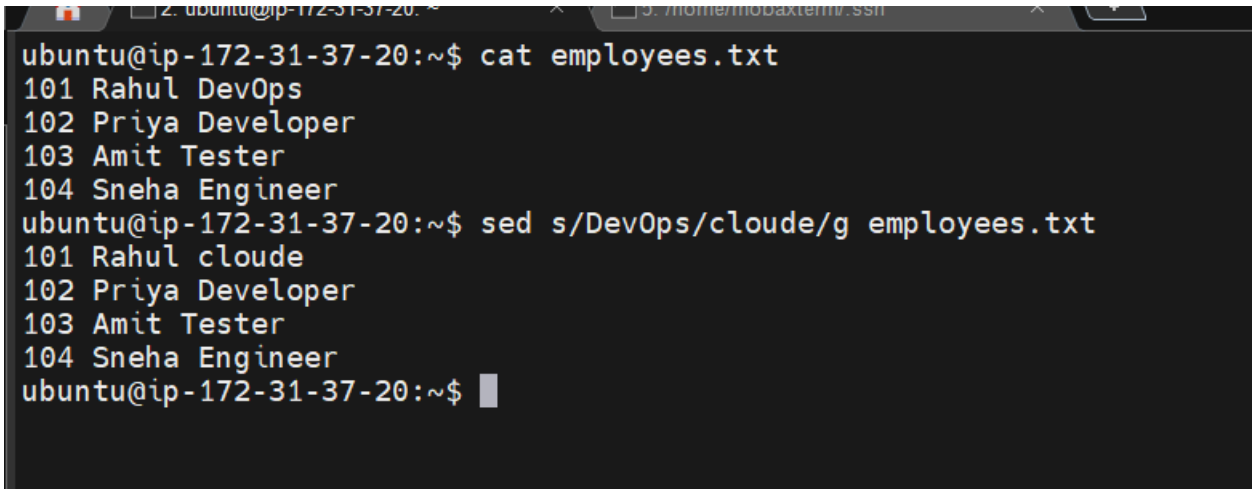
35. Sort lines based on a specific field or column.



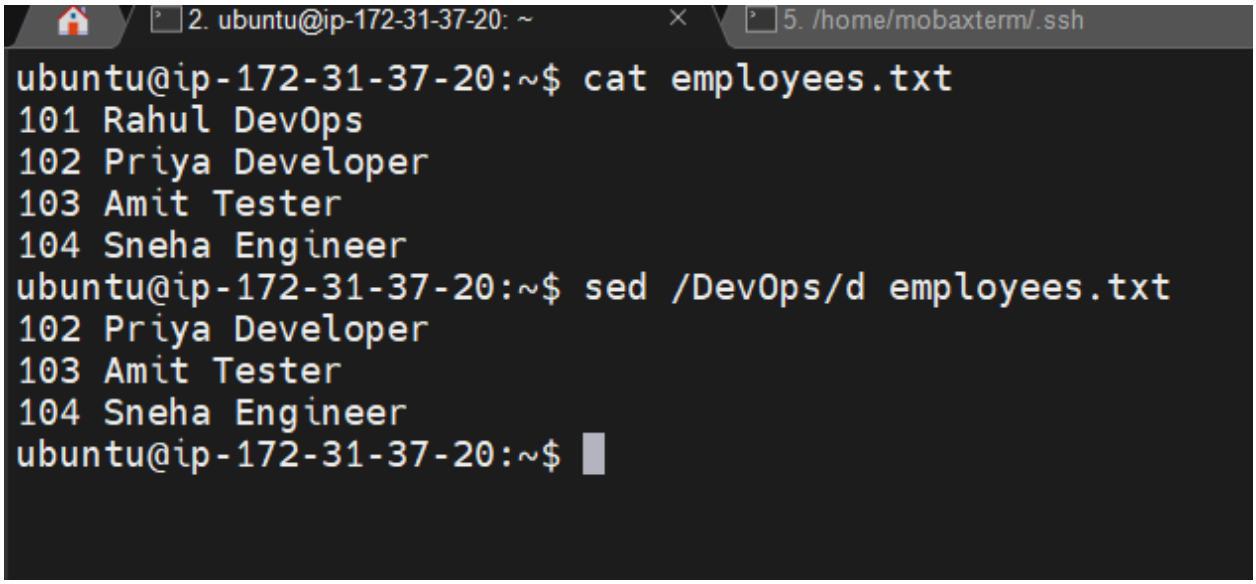
36. Merge multiple files based on a common field or column.



37. Substitute text in a file using search and replace.



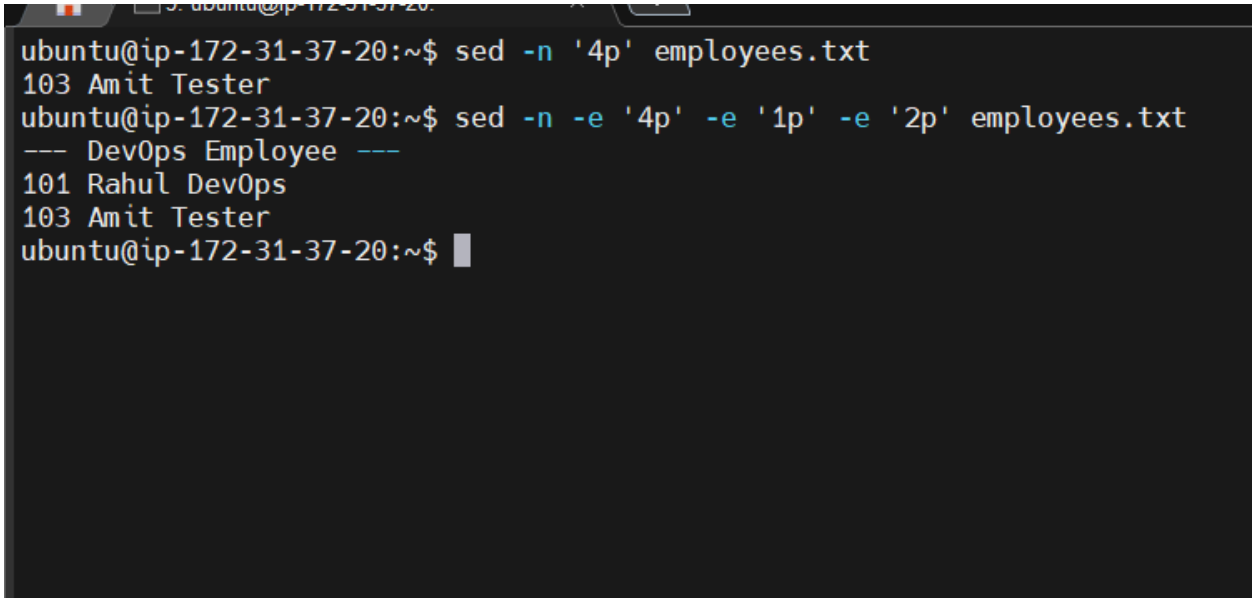
38. Delete specific lines based on a pattern or line number.



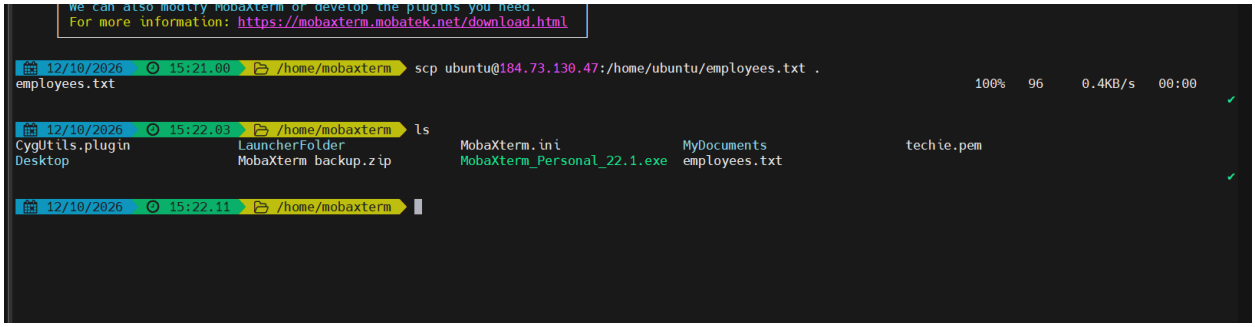
39. Append or insert text before or after a specific pattern or line.

```
ubuntu@ip-172-31-37-20:~$ ls
employees.txt  salaries.txt
ubuntu@ip-172-31-37-20:~$ cat employees.txt
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ sed '/DevOps/i\--- DevOps Employee ---' employees.txt
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ cat employees.txt
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ sed -i '/DevOps/i\--- DevOps Employee ---' employees.txt
ubuntu@ip-172-31-37-20:~$ cat employees.txt
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ sed '/DevOps/a\--- End of DevOps Record ---' employees.txt
--- DevOps Employee ---
--- End of DevOps Record ---
101 Rahul DevOps
--- End of DevOps Record ---
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$
```

40. Print only specific lines from a file.

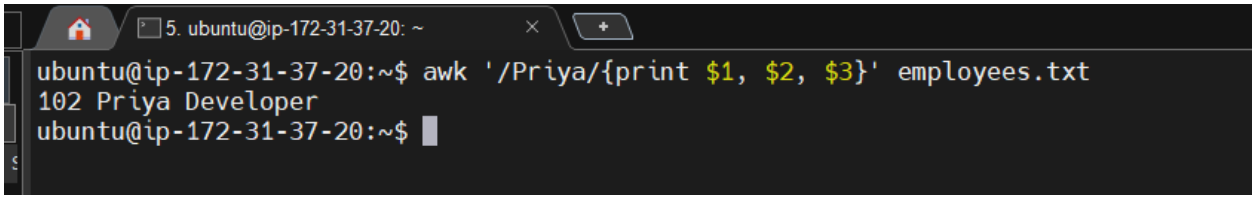


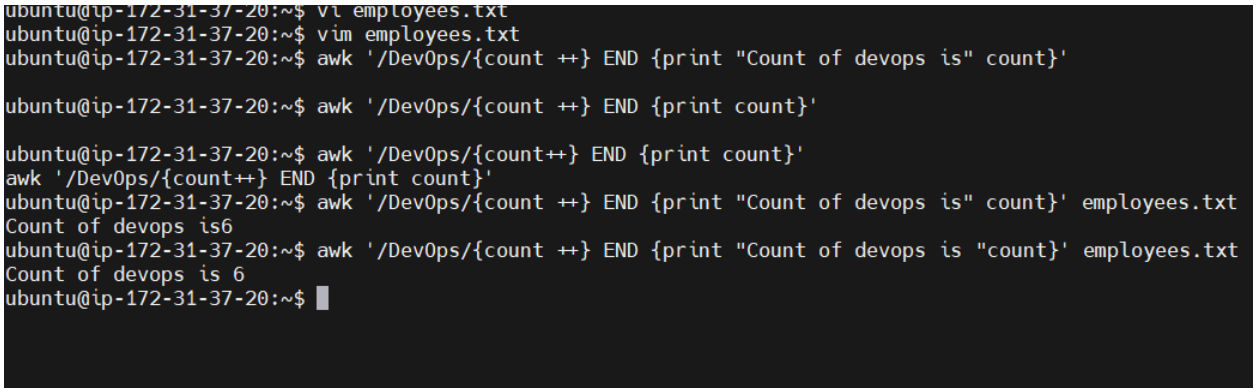
41. Copy file from linux to windows machine

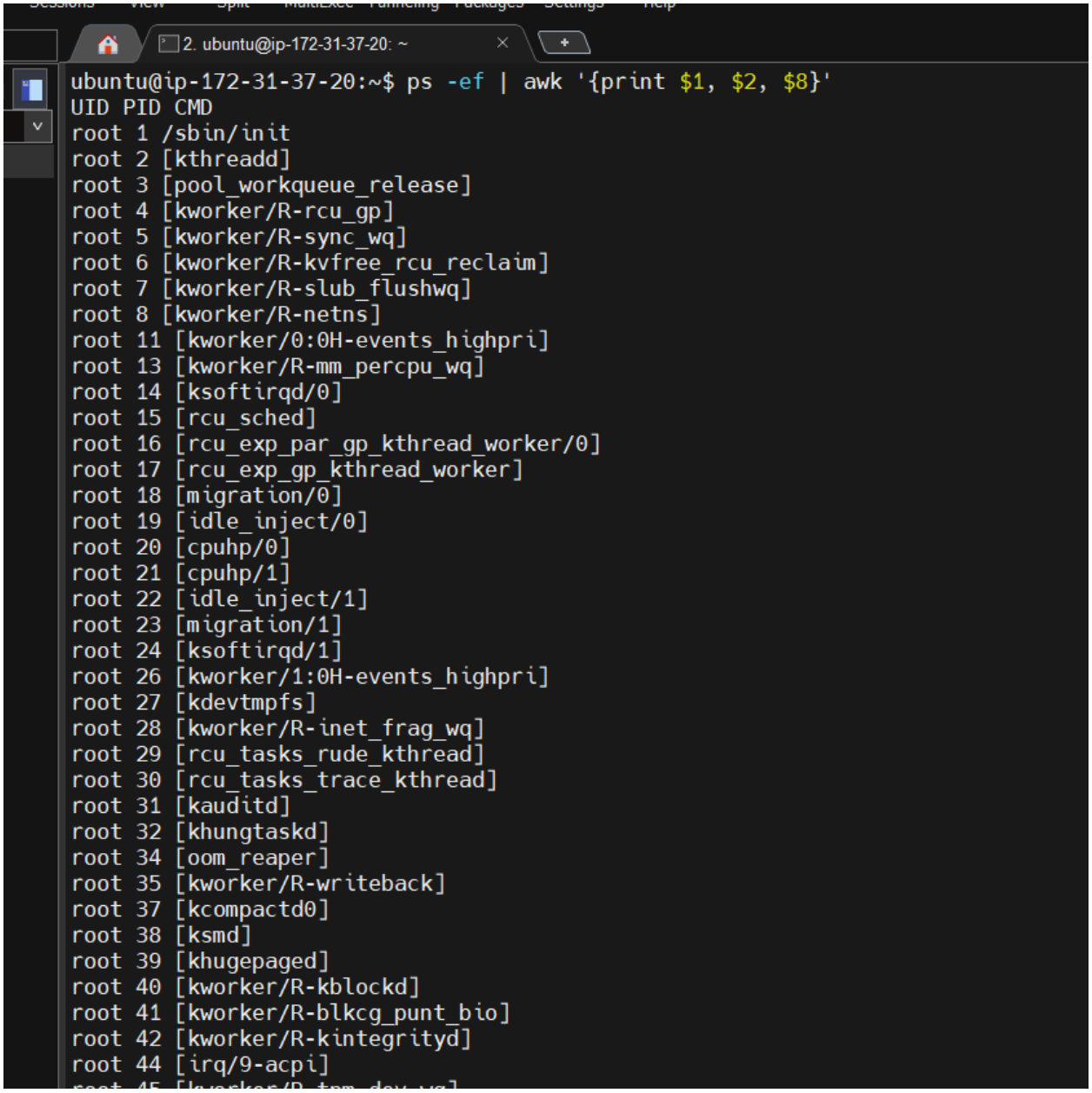


42. 5 use cases for AWK and 5 use cases for sed


5. ubuntu@ip-172-31-37-20: ~
ubuntu@ip-172-31-37-20:~$ awk '{print $1,$2}' employees.txt
--- DevOps
101 Rahul
102 Priya
103 Amit
104 Sneha
ubuntu@ip-172-31-37-20:~$

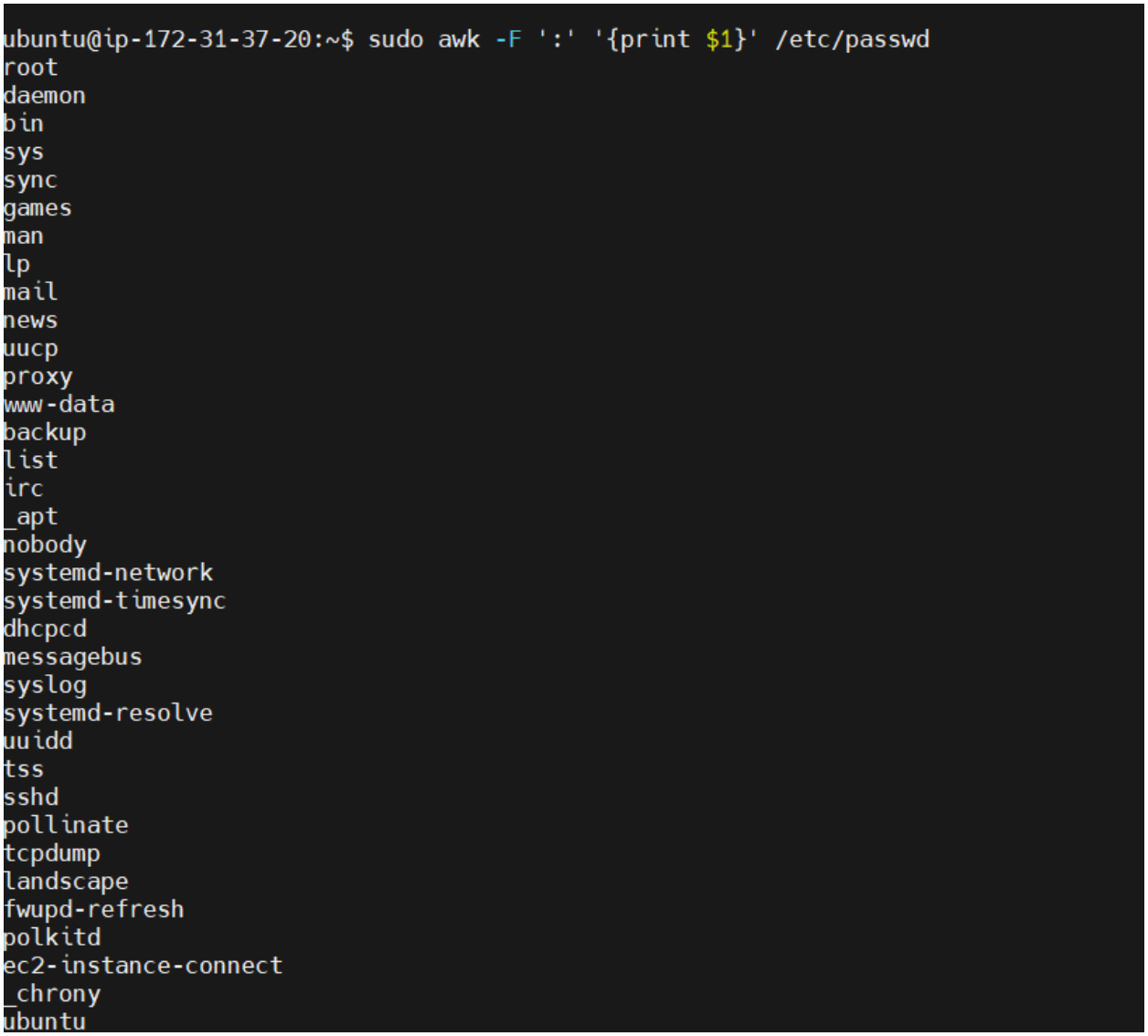

5. ubuntu@ip-172-31-37-20: ~
ubuntu@ip-172-31-37-20:~$ awk '/Priya/{print $1, $2, $3}' employees.txt
102 Priya Developer
ubuntu@ip-172-31-37-20:~$


ubuntu@ip-172-31-37-20:~$ vi employees.txt
ubuntu@ip-172-31-37-20:~$ vim employees.txt
ubuntu@ip-172-31-37-20:~$ awk '/DevOps/{count ++} END {print "Count of devops is" count}'
ubuntu@ip-172-31-37-20:~$ awk '/DevOps/{count ++} END {print count}'
ubuntu@ip-172-31-37-20:~$ awk '/DevOps/{count++} END {print count}'
awk '/DevOps/{count++} END {print count}'
ubuntu@ip-172-31-37-20:~$ awk '/DevOps/{count ++} END {print "Count of devops is" count}' employees.txt
Count of devops is6
ubuntu@ip-172-31-37-20:~$ awk '/DevOps/{count ++} END {print "Count of devops is "count}' employees.txt
Count of devops is 6
ubuntu@ip-172-31-37-20:~$


2. ubuntu@ip-172-31-37-20: ~
ubuntu@ip-172-31-37-20:~$ ps -ef | awk '{print $1, $2, $8}'
UID PID CMD
root 1 /sbin/init
root 2 [kthreadd]
root 3 [pool_workqueue_release]
root 4 [kworker/R-rcu_gp]
root 5 [kworker/R-sync_wq]
root 6 [kworker/R-kvfree_rcu_reclaim]
root 7 [kworker/R-slub_flushwq]
root 8 [kworker/R-netns]
root 11 [kworker/0:0H-events_highpri]
root 13 [kworker/R-mm_percpu_wq]
root 14 [ksoftirqd/0]
root 15 [rcu_sched]
root 16 [rcu_exp_par_gp_kthread_worker/0]
root 17 [rcu_exp_gp_kthread_worker]
root 18 [migration/0]
root 19 [idle_inject/0]
root 20 [cpuhp/0]
root 21 [cpuhp/1]
root 22 [idle_inject/1]
root 23 [migration/1]
root 24 [ksoftirqd/1]
root 26 [kworker/1:0H-events_highpri]
root 27 [kdevtmpfs]
root 28 [kworker/R-inet_frag_wq]
root 29 [rcu_tasks_rude_kthread]
root 30 [rcu_tasks_trace_kthread]
root 31 [kauditd]
root 32 [khungtaskd]
root 34 [oom_reaper]
root 35 [kworker/R-writeback]
root 37 [kcompactd0]
root 38 [ksmd]
root 39 [khugepaged]
root 40 [kworker/R-kblockd]
root 41 [kworker/R-blkcg_punt_bio]
root 42 [kworker/R-kintegrityd]
root 44 [irq/9-acpi]

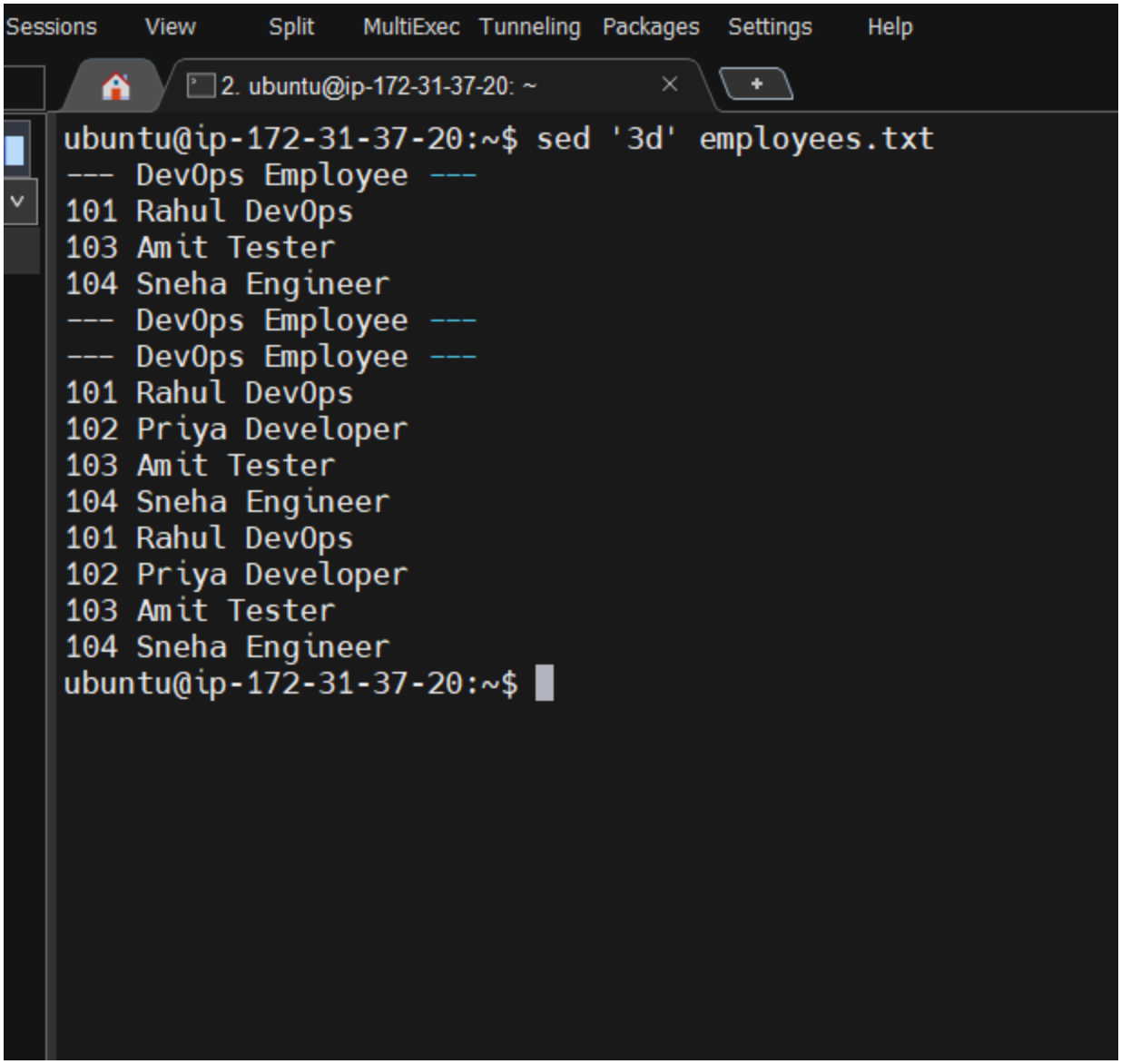
```
ubuntu@ip-172-31-37-20:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        6.8G  3.1G  3.7G  46% /
tmpfs            456M     0  456M   0% /dev/shm
tmpfs            183M  924K  182M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  3.1K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   94M  726M  12% /boot
/dev/nvme0n1p15  105M  6.2M   99M   6% /boot/efi
tmpfs             92M   12K   92M   1% /run/user/1000
ubuntu@ip-172-31-37-20:~$ df -h | awk '{print $1, $4}'
Filesystem Avail
/dev/root 3.7G
tmpfs 456M
tmpfs 182M
tmpfs 5.0M
efivarfs 120K
/dev/nvme0n1p16 726M
/dev/nvme0n1p15 99M
tmpfs 92M
ubuntu@ip-172-31-37-20:~$
```

```
ions  View  Split  MultiExec  Tunneling  Packages  Settings  Help
2. ubuntu@ip-172-31-37-20: ~
ubuntu@ip-172-31-37-20:~$ awk 'NR ⩾ 2 && NR <
>
ubuntu@ip-172-31-37-20:~$ awk 'NR ⩾ 2 && NR ⩽ 4 {print}' employees.txt
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
ubuntu@ip-172-31-37-20:~$
```


ubuntu@ip-172-31-37-20:~$ sudo awk -F ':' '{print $1}' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
syslog
systemd-resolve
uuidd
tss
sshd
pollinate
tcpdump
landscape
fwupd-refresh
polkitd
ec2-instance-connect
_chrony
ubuntu

SED

```
ubuntu@ip-172-31-37-20:~$ cat employees.txt
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
--- DevOps Employee ---
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ sed 's/DevOps/Cloude/g' employees.txt
--- Cloude Employee ---
101 Rahul Cloude
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
--- Cloude Employee ---
--- Cloude Employee ---
101 Rahul Cloude
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
101 Rahul Cloude
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$
```


Sessions View Split MultiExec Tunneling Packages Settings Help
2. ubuntu@ip-172-31-37-20: ~
ubuntu@ip-172-31-37-20:~$ sed '3d' employees.txt
--- DevOps Employee ---
101 Rahul DevOps
103 Amit Tester
104 Sneha Engineer
--- DevOps Employee ---
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$

```
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$ sed '/^$/d' employees.txt
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
--- DevOps Employee ---
--- DevOps Employee ---
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
101 Rahul DevOps
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
ubuntu@ip-172-31-37-20:~$
```

```
ubuntu@ip-172-31-37-20:~$ sed -n '3,7p' employees.txt
102 Priya Developer
103 Amit Tester
104 Sneha Engineer
--- DevOps Employee ---
--- DevOps Employee ---
ubuntu@ip-172-31-37-20:~$
```